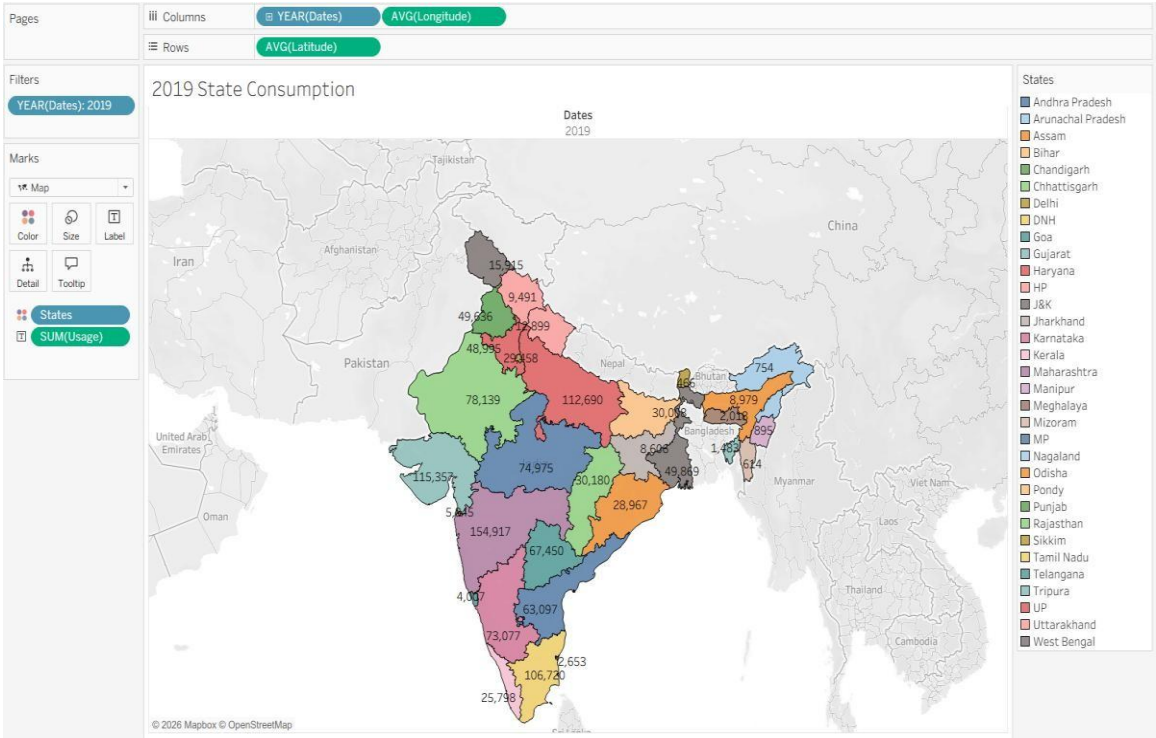
Plugging into Future Project

Preprocessing Steps

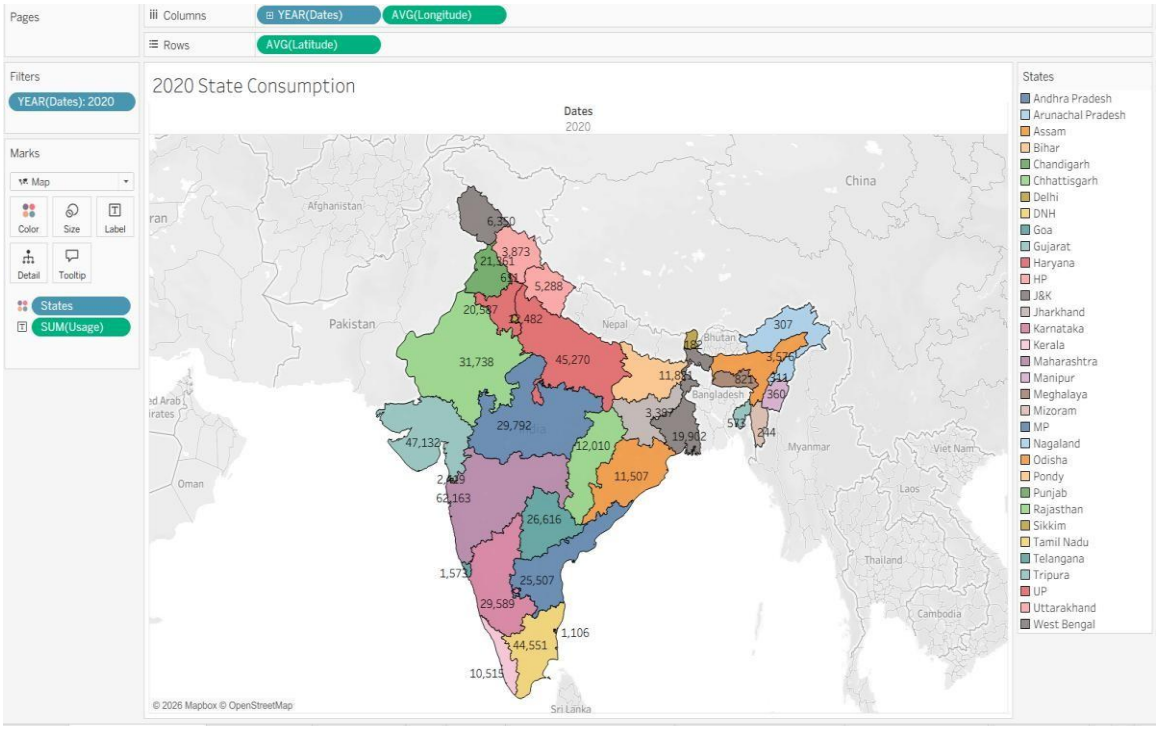
1. Imported electricity consumption dataset (CSV format).
2. Checked and handled missing/null values.
3. Standardized column names and data formats.
4. Converted year and month fields into proper date formats.
5. Aggregated data using SUM and GROUPBY (state, region, year, quarter).
6. Created calculated fields (Total Consumption, Regional Totals, Top/Bottom Rankings).
7. Filtered dataset for specific years (2019, 2020).
8. Separated pre-lockdown and during-lockdown data for comparison.
9. Prepared data for visualization-friendly structure (JSON format via Flask).

Business Questions with Visualization

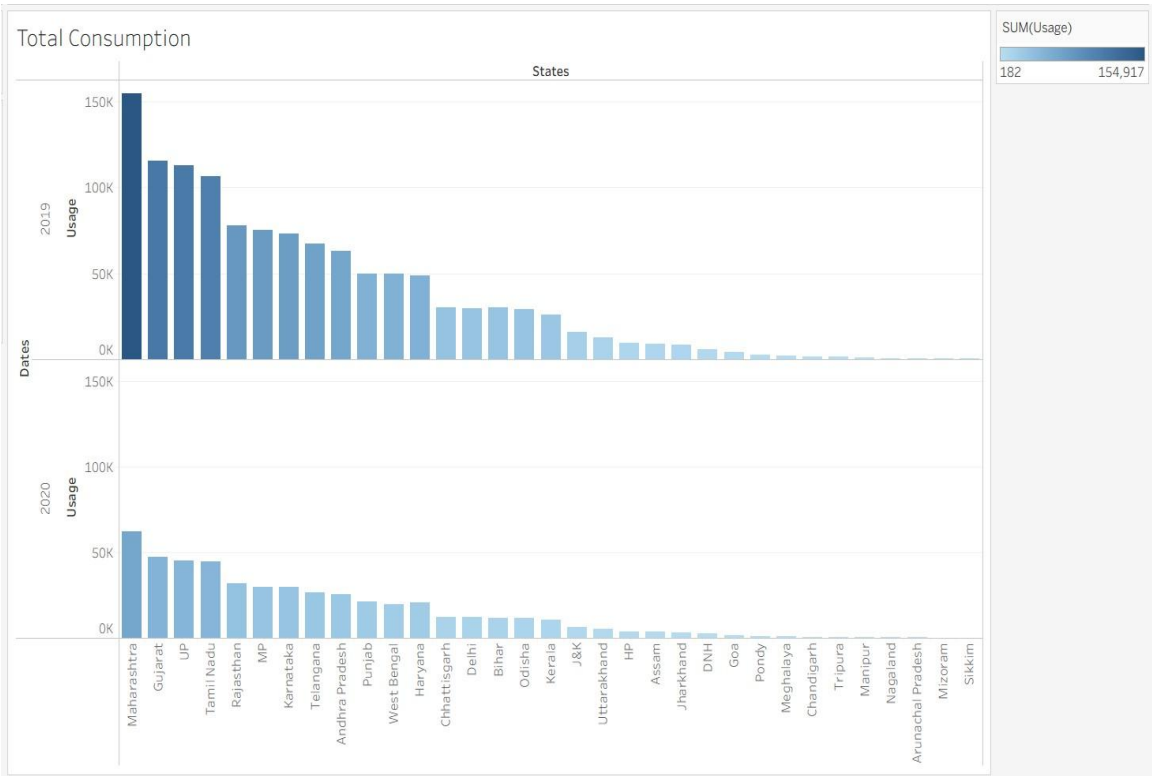
1. What was the state-wise electricity consumption in 2019?



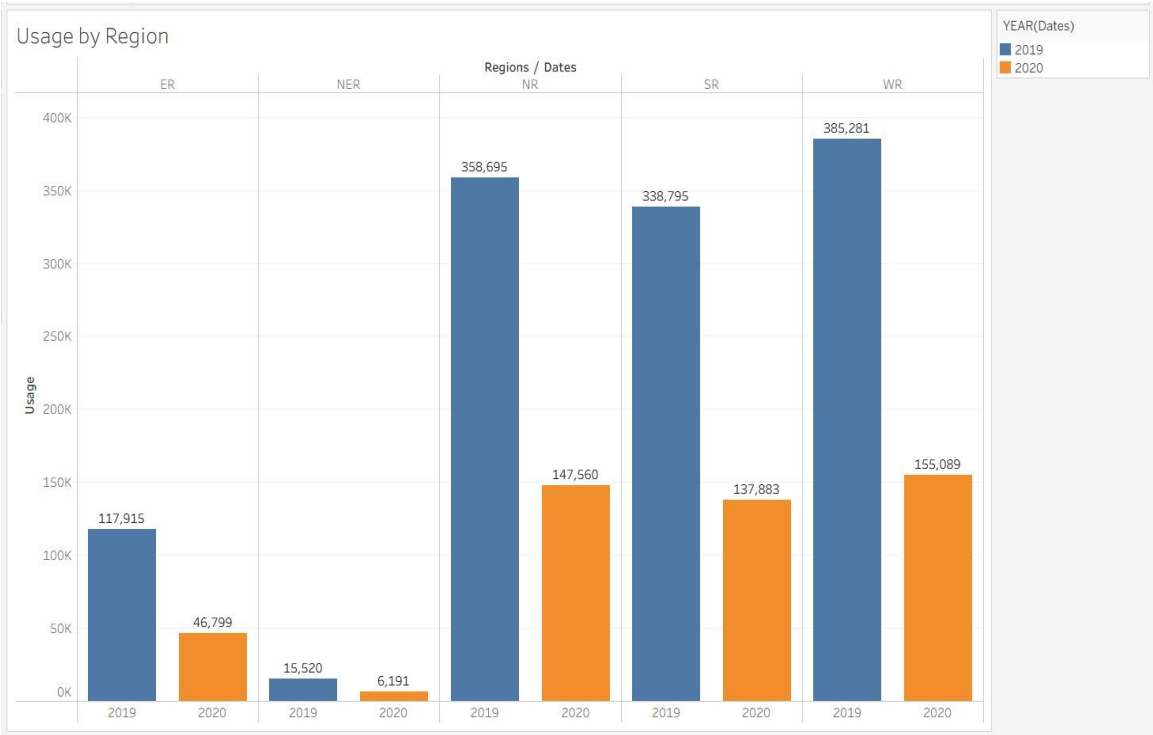
2. What was the state-wise electricity consumption in 2020?



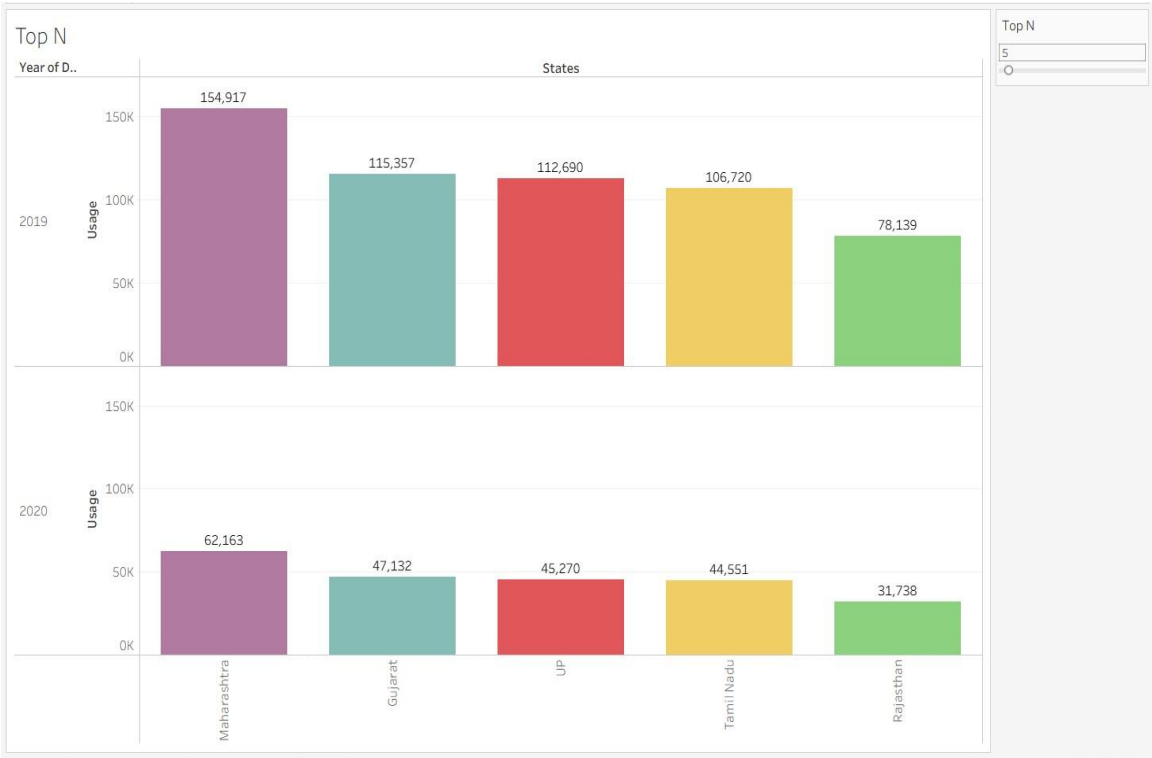
3. What is the overall total electricity consumption?



4. How does electricity usage vary by region?

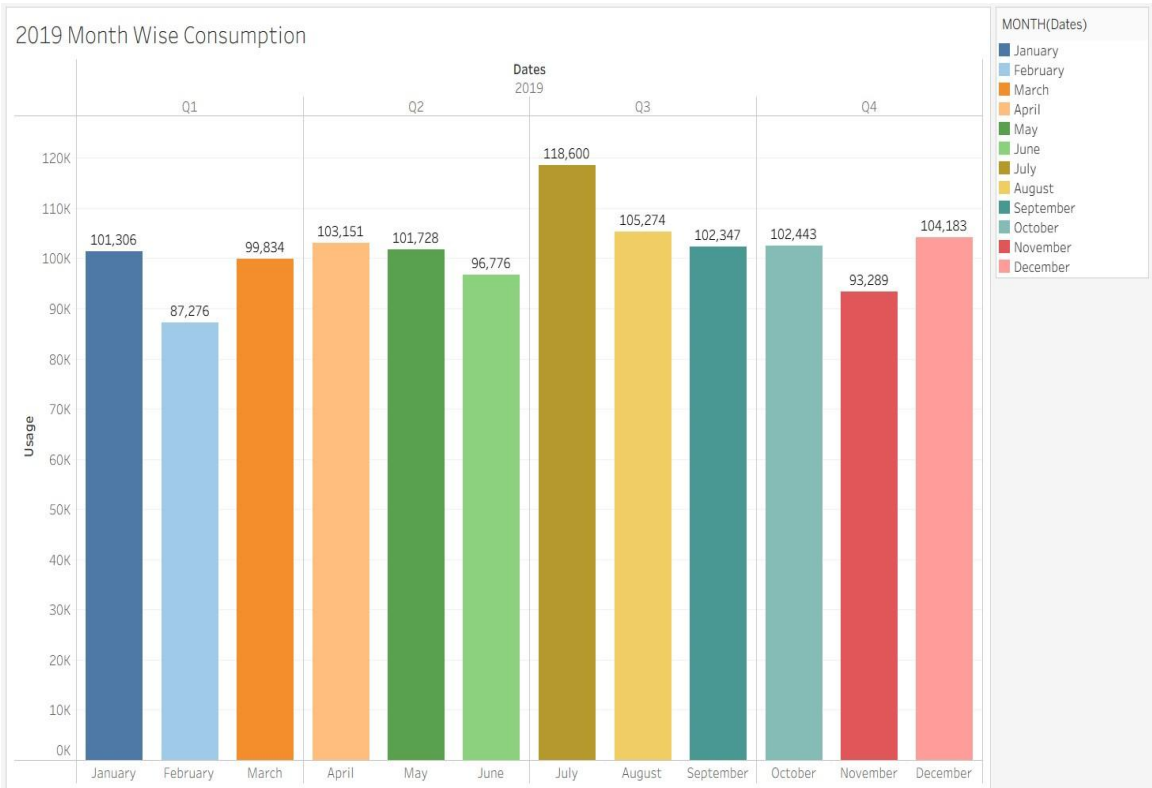


5. Which states have the highest electricity consumption?

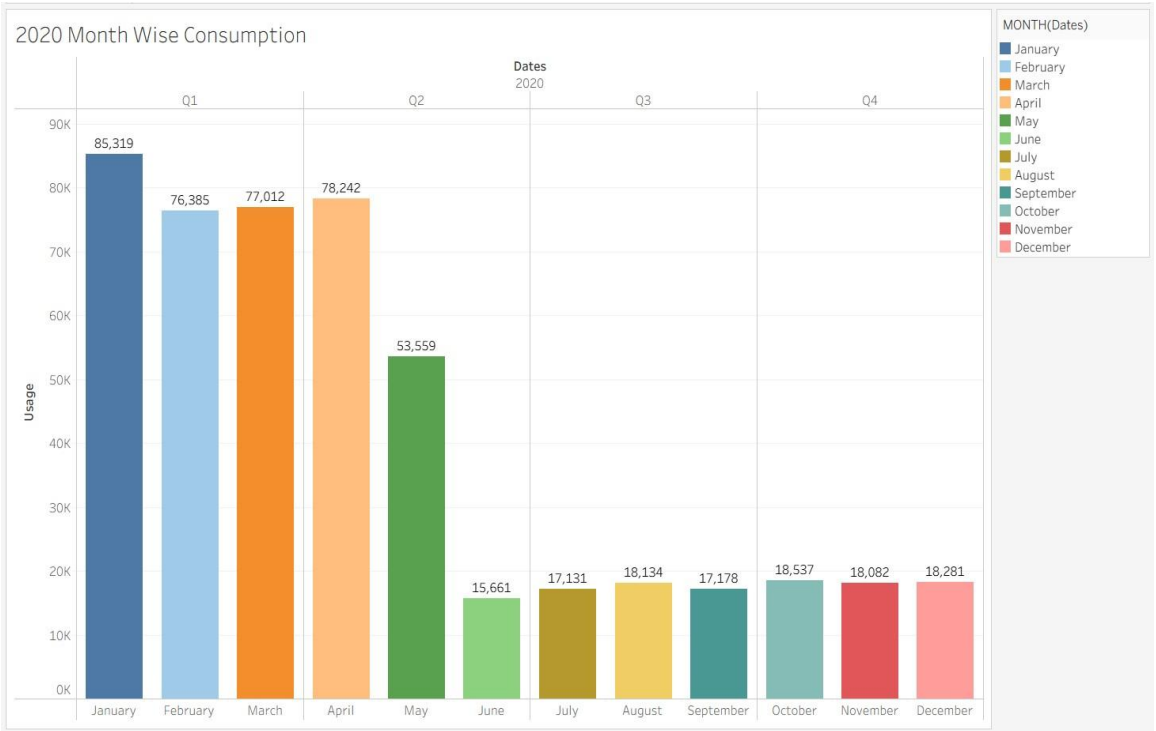


6. Which states have the lowest electricity consumption?

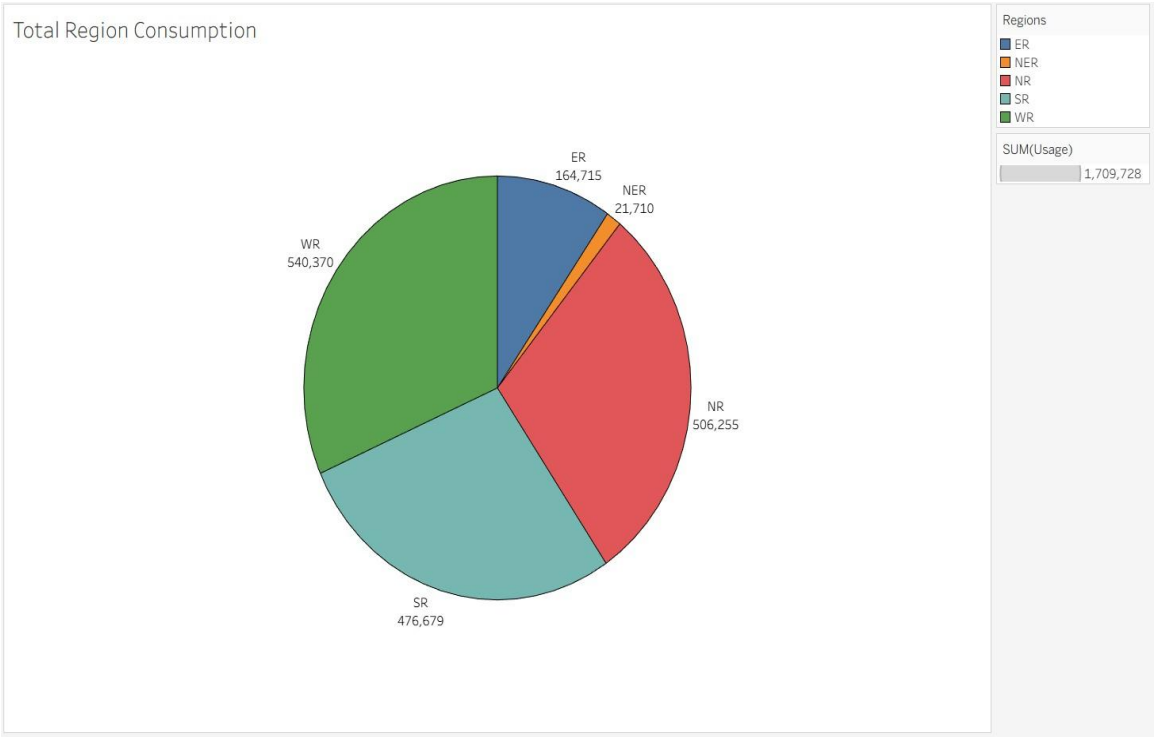
7. What was the month-wise consumption in 2019?



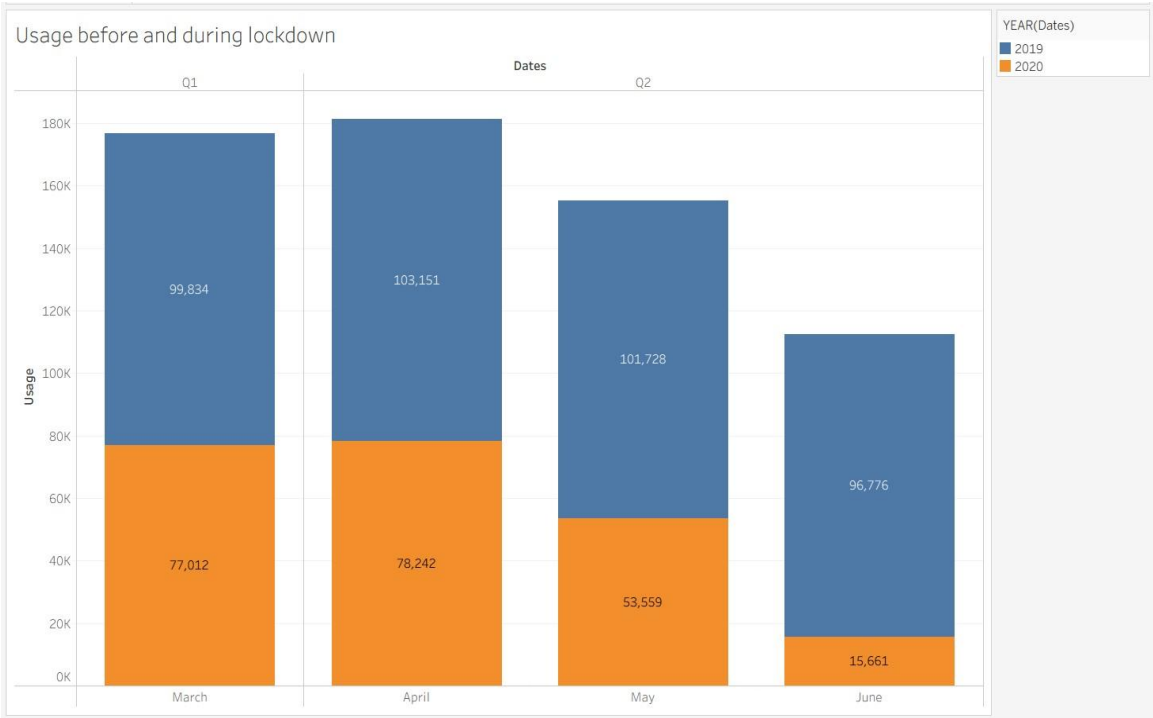
8. What was the month-wise consumption in 2020?



9. What is the total consumption by region?

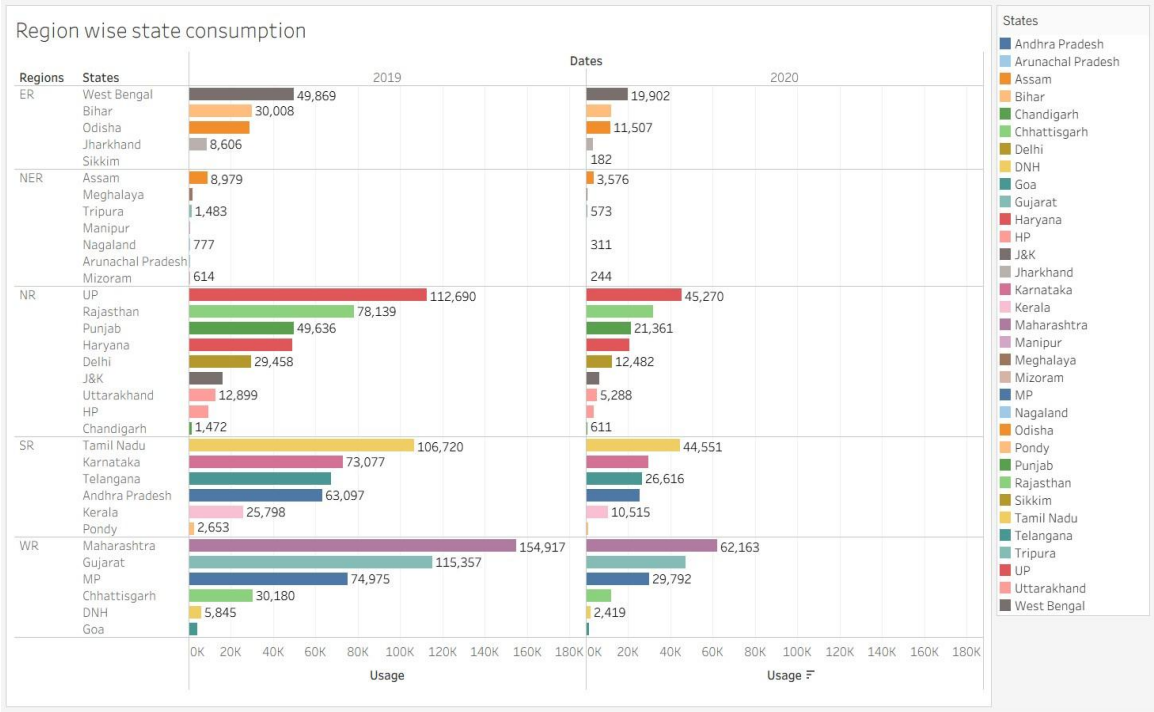


10. How did electricity usage change before and during lockdown?

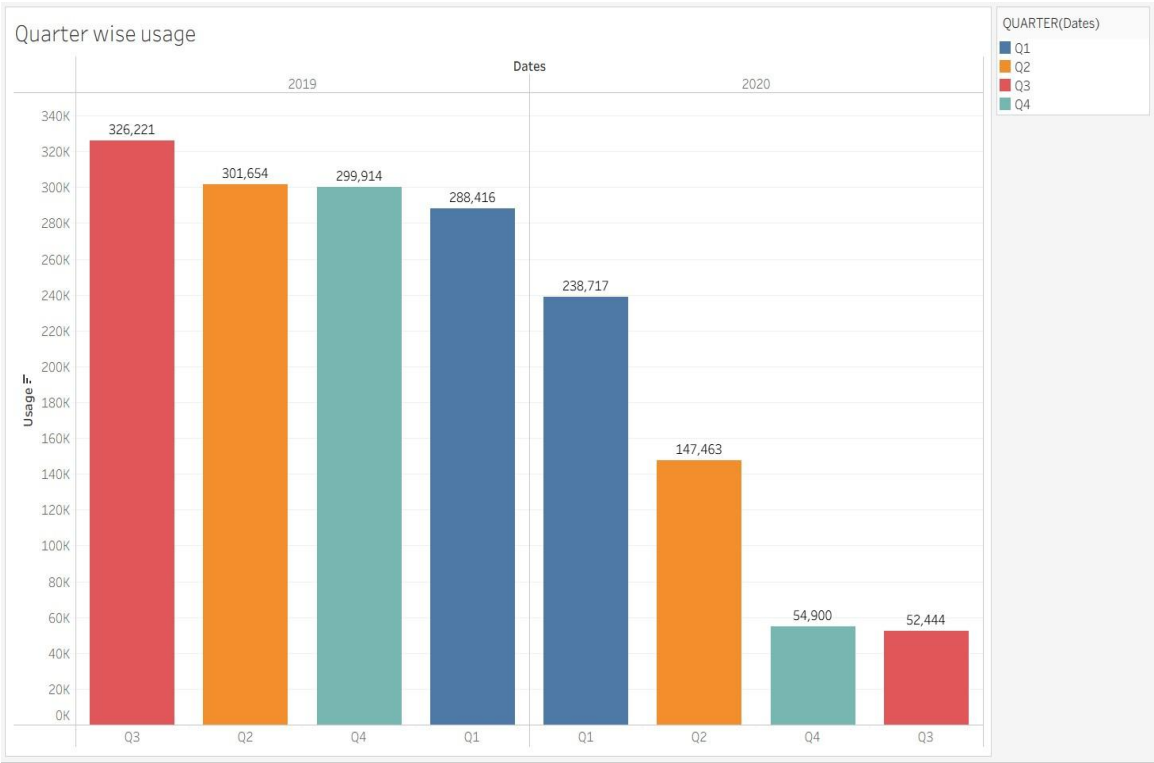


11. Which month recorded the highest consumption?

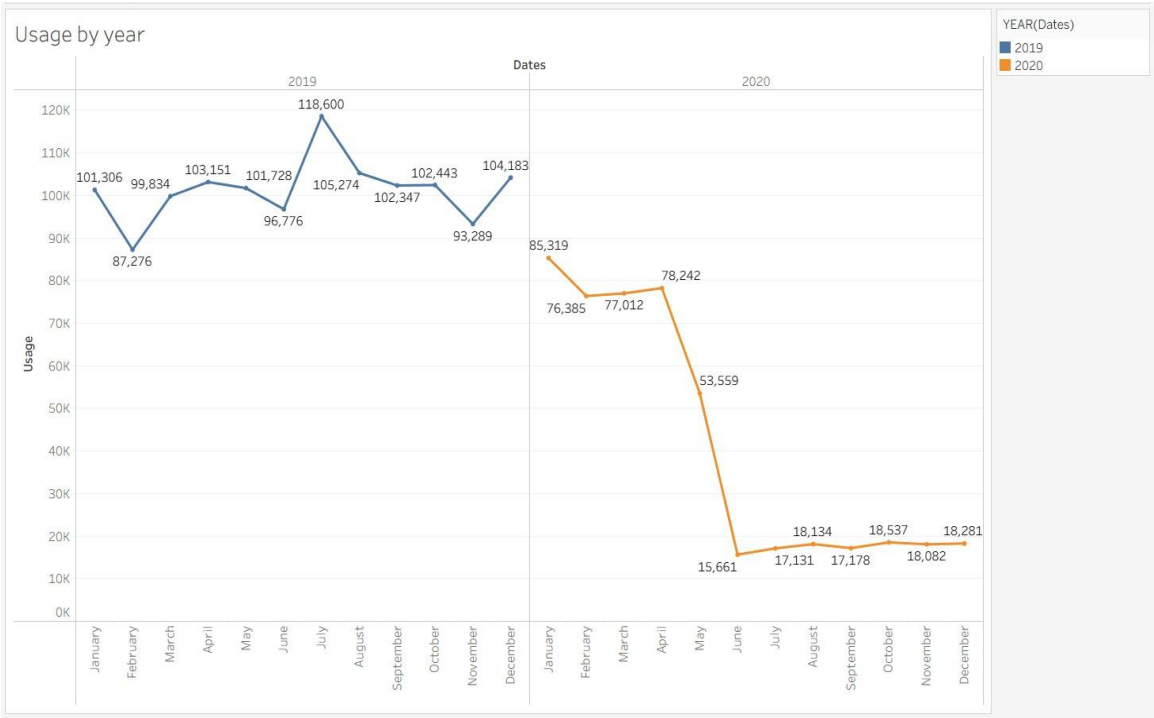
12. How does consumption vary state-wise within each region?



13. What is the quarter-wise electricity usage?

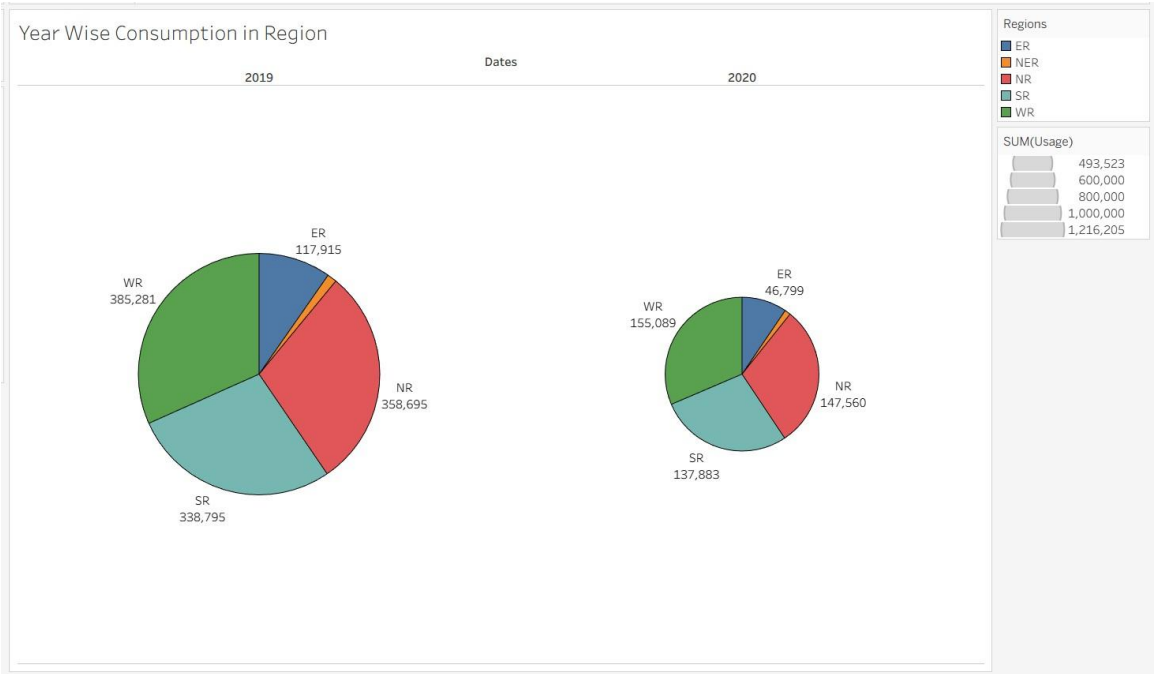


14. How has electricity consumption changed year-wise?



15. Which metro city states consume more electricity?

16. How does electricity consumption change year-wise across regions?


Year Wise Consumption in Region
Dates
2019
2020
ER
117,915
NR
358,695
SR
338,795
WR
385,281
ER
46,799
NR
147,560
SR
137,883
WR
155,089
Regions
ER
NER
NR
SR
WR
SUM(Usage)
493,523
600,000
800,000
1,000,000
1,216,205